

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1095

Chatom 138 kV
July 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1095. The Interconnection Customer proposes a 62.6 MW wind facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on October 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,247,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Purdy Substation – Luray Hydro Sta 138.0 kV
Contingency: Chatom – Purdy 138.0 kV
Loading: 135.8% (Rating: 210 MVA)
DFAX: 7.8%
Pre-project loading: 133.5%
2. Facility: Puddledock Rd – Luray Hydro Sta 138.0 kV L
Contingency: Chatom – Purdy 138.0 kV L
Loading: 135.5% (Rating: 210 MVA)
DFAX: 7.8%
Pre-project loading: 133.2%
3. Facility: St Louis – Nichols 138.0 kV L
Contingency: Page – Purdy 138.0 kV
Loading: 128.9% (Rating: 346 MVA)
DFAX: 20.3%
Pre-project loading: 125.2%
4. Facility: Elkton D.P – Roosevelt Rd 138.0 kV line
Contingency: St Louis – Nuclear Fuels 138.0 kV
Loading: 126.3% (Rating: 251 MVA)
DFAX: 12.5%
Pre-project loading: 123.2%
5. Facility: Elkton D.P – Houston 138.0 kV
Contingency: St Louis – Nuclear Fuels 138.0 kV
Loading: 126.3% (Rating: 251 MVA)
DFAX: 12.5%
Pre-project loading: 123.2%

6. Facility: S Belleville – Nichols 138.0 kV
Contingency: Page Substation – Purdy 138.0 kV
Loading: 125.2% (Rating: 352 MVA)
DFAX: 20.3%
Pre-project loading: 121.6%
7. Facility: Roosevelt Rd – Wadsworth 138.0 kV
Contingency: St Louis – Nuclear Fuels Substation 138.0 kV
Loading: 124.9% (Rating: 260 MVA)
DFAX: 12.5%
Pre-project loading: 121.9%
8. Facility: Puddledock Rd – Mount Jackson 138.0 kV
Contingency: Purdy Substation – Chatom 138.0 kV
Loading: 121.1% (Rating: 240 MVA)
DFAX: 7.8%
Pre-project loading: 119.0%
9. Facility: Siloam Springs – Detour 138.0 kV line
Contingency: Detour – Ft Benton 138.0 kV
Loading: 120.8% (Rating: 253 MVA)
DFAX: 14.7%
Pre-project loading: 117.2%
10. Facility: Detour – Cedarville Junction 138.0 kV
Contingency: St Louis – Nichols 138.0 kV
Loading: 119.6% (Rating: 411 MVA)
DFAX: 22.0%
Pre-project loading: 116.3%
11. Facility: Cedarville Jct – Page 138.0 kV
Contingency: St Louis – Nichols 138.0 kV
Loading: 119.3% (Rating: 416 MVA)
DFAX: 22.0%
Pre-project loading: 116.0%
12. Facility: Page – Purdy 138.0 kV Line
Contingency: St Louis – Nichols 138.0 kV
Loading: 118.2% (Rating: 431 MVA)
DFAX: 22.0%
Pre-project loading: 115.0%
13. Facility: Detour – Fort Benton 138.0 kV
Contingency: Siloam Springs – Detour Substation 138.0 kV
Loading: 117.0% (Rating: 315 MVA)
DFAX: 15.8%
Pre-project loading: 113.8%
14. Facility: Springfield Power Llc – Edinburg 138.0 kV
Contingency: American – Urban 345.0 kV
Loading: 114.9% (Rating: 175 MVA)
DFAX: 9.6%
Pre-project loading: 111.5%
15. Facility: Video Pd – Mcauley 138.0 kV
Contingency: Declo – New London 138.0 kV L
Loading: 109.8% (Rating: 228 MVA)
DFAX: 5.3%
Pre-project loading: 108.3%
16. Facility: S Jordan – Mount Pleasant Substation 138.0 kV
Contingency: Firebantz – Mount Pleasant 138.0 kV

Loading: 107.4% (Rating: 188 MVA)

DFAX: 6.3%

Pre-project loading: 105.3%

17. Facility: Roslyn – S Bluefield 138.0 kV L

Contingency: Urban – American 345.0 kV

Loading: 107.0% (Rating: 211 MVA)

DFAX: 9.6%

Pre-project loading: 104.1%

18. Facility: Sprigfield Power Llc – S Blefield 138.0 kV Line

Contingency: Urban Substation – American 345.0 kV

Loading: 106.4% (Rating: 206 MVA)

DFAX: 9.6%

Pre-project loading: 103.4%

19. Facility: Lewisburg – Siloam Springs 138.0 kV

Contingency: Detour – Ft Benton 138.0 kV

Loading: 100.9% (Rating: 307 MVA)

DFAX: 6.2%

Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Purdy Substation – Luray Hydro Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$2,603,000	28% (\$717,000)
Puddledock Rd – Luray Hydro Sta 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,226,000	28% (\$1,439,000)
St Louis – Nichols 138.0 kV	Rebuild line with higher-capacity conductor	\$6,463,000	9% (\$602,000)
Elkton D.P – Roosevelt Rd 138.0 kV line	Rebuild line with higher-capacity conductor	\$979,000	8% (\$81,000)
Houston – Elkton D.P Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,772,000	8% (\$229,000)
S Belleville – Nichols 138.0 kV	Rebuild line with higher-capacity conductor	\$7,708,000	9% (\$718,000)
Roosevelt Rd – Wadsworth 138.0 kV	Rebuild line with higher-capacity conductor	\$5,188,000	8% (\$428,000)
Puddledock Rd – Mount Jackson 138.0 kV	Rebuild line with higher-capacity conductor	\$5,577,000	28% (\$1,536,000)
Siloam Springs – Detour 138.0 kV	Rebuild line with higher-capacity conductor	\$15,495,000	29% (\$4,543,000)
Detour – Cedarville Junction 138.0 kV	Rebuild line with higher-capacity conductor	\$9,326,000	21% (\$1,921,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cedarville Jct – Page 138.0 kV	Rebuild line with higher-capacity conductor	\$10,829,000	21% (\$2,230,000)
Page – Purdy 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,270,000	21% (\$674,000)
Detour – Fort Benton 138.0 kV	Rebuild line with higher-capacity conductor	\$8,027,000	29% (\$2,343,000)
Springfield Power Llc – Edinburg 138.0 kV	Rebuild line with higher-capacity conductor	\$7,118,000	27% (\$1,935,000)
Video Pd – Mcauley 138.0 kV	Reconductor line	\$892,000	16% (\$142,000)
South Jordan – Mount Pleasant 138.0 kV	Reconductor line	\$2,429,000	47% (\$1,147,000)
Roslyn – S Bluefield 138.0 kV	Reconductor line	\$3,097,000	27% (\$842,000)
Sprigfield Power Llc – S Blefield 138.0 kV Line	Reconductor line	\$1,762,000	27% (\$479,000)
Lewisburg – Siloam Springs 138.0 kV	Reconductor line	\$2,617,000	47% (\$1,241,000)
	Total allocated to project		\$23,247,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.